

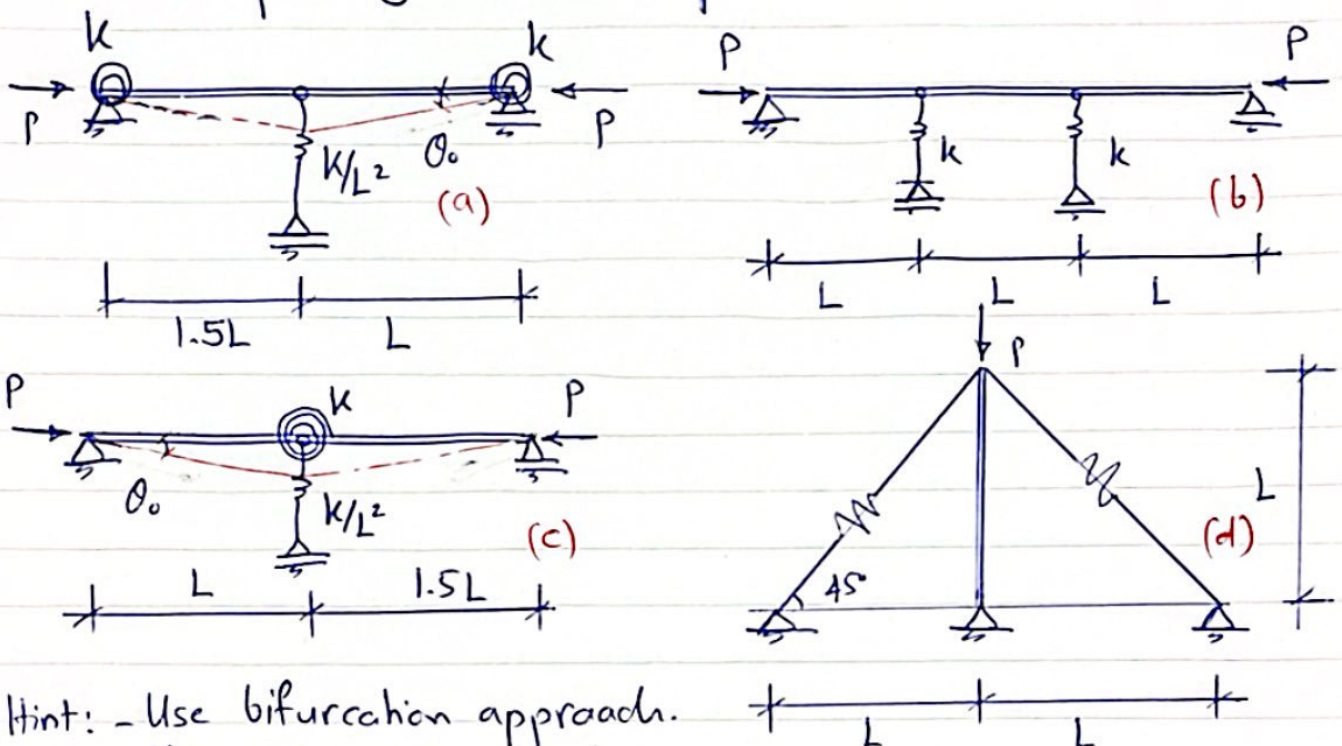
Assignment * 1

STR 655

Spring 2026

1. Read chapter (1) in text book.

2. Using small deflection analysis compute (P_{cr}) and corresponding buckled shape for:



Hint: - Use bifurcation approach.
- Use energy approach

3. Using large deflection theory for problems (2a) & (2c) determine:-

- post-buckling path
- plot fundamental versus post-buckling path.
- check type of equilibrium along:
 - fundamental path
 - post buckling path.
 - critical load.
- Investigate effect of geometric imperfection